

All orders refinement for unlabeled permutation graphs

Factorial expansion, inverse thresholds, and sampling in OEIS A123448

Report 153

3 October 2026

Abstract

Let a_n count unlabeled permutation graphs on n vertices. We establish, for every fixed $R \geq 1$,

$$a_n = \frac{1}{4} \sum_{k=0}^{R-1} h_k(n-k)! + O_R((n-R)!),$$

with a formal coefficient generator using only finite unrooted and vertex-rooted graph counts at each order. The first coefficients are 1, -4 , -1 , $-94/3$, $-769/6$, $-19969/15$, $-531812/45$, and $-40114096/315$. The main step is a finite-polynomial approximation: all but $O_R((n-R)!)$ permutation realizers have a recoverable large prime quotient, bounded decorations, and a bounded vertex-rooted context. Borinsky’s established factorial composition theorem then transfers all fixed orders without assuming an asymptotic expansion of the unknown graph series. We derive denominator divisibility, arbitrary fixed-order threshold brackets, an explicit second Lambert- W inverse shift, and the sharp all-orders sampling comparison beginning $d_{TV} = 4/n - 15/n^2 + O(n^{-3})$ between uniform graph classes and graphs of uniform permutations. Exact companion programs provide independent coefficient routes, formal inverse checks, and two exhaustive finite enumeration options.

Scope and priority. The exact recurrence and table in Bayoumi–El-Zahar–Khamis (1990), classical two-dimensional-poset asymptotics and sampling precedents, and the simple-permutation asymptotic machinery are prior art. This report presents an all-orders refinement; it makes no first-ever claim for the leading equivalent. The final 2026 intersection-graph article and two older subscription articles were identified but their full final texts were not inspected. All truncation orders are fixed; no effective onset or growing-order remainder estimate is claimed.

[write] **Status and trust boundary, added 5 October 2026, batch 106.** This is bundle Report 153 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a123448-permutation-graphs** (provenance, the sources as the write read them, relation to the repository, reading conventions and the collected non-claims in Section 1.1). It is unrefereed and not formalized; its author line reads “Report 153” and its PDF author field is empty. *What rests on what.* Theorem 1.1 has a conventional proof that uses four external inputs without re-proving them: Borinsky’s factorial composition and inversion theorem [3, Theorem 35, equations (62)–(67)], the Albert–Atkinson–Klazar decomposition and simple-permutation equation [1], the Habib–Paul identification of strong modules with strong common intervals [4, Lemma 20], and the classical fact that a prime permutation graph and its complement each have exactly two transitive orientations (Section 2.2). The eight values $h_0, \dots, h_7$ use in addition the finite counts $a_1, \dots, a_9$ and $r_1, \dots, r_8$, enumerated exhaustively by two programs (Section 7); the sampling coefficients use the counts $b_j, c_j, j \leq 8$. No asymptotic statement rests on a computation, and no constant or onset is effective. The write adds Remark 8.2 (the inverse of Section 8 in the terms of the repository’s transseries volume), dated notes on h_8 (end of Section 7), on the 1990 table, on Johnston’s question and on the repository (Section 10), and Section 12.1, the open questions.

Contents

1	Results and the proof mechanism	3
2	Structural conventions and established inputs	6
3	Excluding middle intervals	8
4	Recovering one large prime node	9
5	Burnside weights and negligible symmetric quotients	11
6	Factorial transfer and all orders	12
7	Exact coefficients and finite certification	13
8	Threshold inversion with rounding preserved	15
9	All orders comparison of the sampling laws	19
10	Source overlap and the boundary of the contribution	22
11	Reproducibility and verification limits	25
12	Further research questions	25

1 Results and the proof mechanism

A permutation $\pi \in \mathcal{S}_n$ determines an inversion graph on $\{1, \dots, n\}$: vertices $i < j$ are adjacent exactly when $\pi(i) > \pi(j)$. A *permutation graph* is an abstract graph isomorphic to such a graph. Throughout, graphs are simple and unlabeled; a graph and its complement are counted separately unless they happen to be isomorphic. Write a_n for the number of isomorphism classes on n vertices, and r_n for the number of classes with one distinguished vertex, with isomorphisms required to preserve that vertex. Thus r_n counts vertex orbits across unrooted graph classes, not na_n .

Introduce ordinary formal power series

$$A(x) = \sum_{n \geq 1} a_n x^n, \quad C(x) = \sum_{d \geq 0} r_{d+1} x^d, \quad F(x) = \sum_{n \geq 1} n! x^n. \quad (1.1)$$

Let $T = F^{(-1)}$ denote formal compositional inversion, and put $Q = T \circ A$. Neither convergence nor a positive radius of convergence is intended for these series.

Theorem 1.1 (All fixed orders). *Define*

$$H(x) = C(x) \frac{xQ'(x)}{Q(x)A'(x)} \exp\left(\frac{1}{x} - \frac{1}{Q(x)}\right) = \sum_{k \geq 0} h_k x^k. \quad (1.2)$$

This is a well-defined series in $\mathbb{Q}[[x]]$, with $H(0) = 1$. For each fixed integer $R \geq 1$, as $n \rightarrow \infty$,

$$a_n = \frac{1}{4} \sum_{k=0}^{R-1} h_k (n-k)! + O_R((n-R)!). \quad (1.3)$$

The coefficient h_k depends only on $a_1, \dots, a_{k+2}$ and $r_1, \dots, r_{k+1}$. These are finite, exactly enumerable inputs. Moreover, $k!h_k \in \mathbb{Z}$ for every $k \geq 0$.

The first eight h_k are displayed in Section 7. In particular,

$$a_n = \frac{n!}{4} \left(1 - \frac{4}{n} - \frac{1}{n^2} - \frac{97}{3n^3} + O(n^{-4})\right). \quad (1.4)$$

The inverse-factorial and inverse-power coefficient lists are *different*: the $-94/3$ coefficient multiplies $(n-3)!$ in (1.3), whereas $-97/3$ multiplies n^{-3} after division by $n!$.

The leading equivalent gives a natural comparison of sampling laws. Let $\mathbf{P}_n$ be the law of the unlabeled inversion graph of a uniform permutation in $\mathcal{S}_n$, and let $\mathbf{U}_n$ be uniform on the a_n graph classes. With $d_{\text{TV}}(\mu, \nu) = \sup_E |\mu(E) - \nu(E)|$, we prove

$$d_{\text{TV}}(\mathbf{P}_n, \mathbf{U}_n) = \frac{4}{n} - \frac{15}{n^2} + \frac{169}{3n^3} + \frac{215}{2n^4} + O(n^{-5}). \quad (1.5)$$

This controls events and uniformly bounded observables. It does not, by itself, transfer expectations of unbounded statistics or relative errors for rare events.

For $N(X) = \min\{n \geq 1 : a_n \geq X\}$, the report gives arbitrary fixed-order smooth approximations and a shrinking rounding window. In particular, with $L = \log X$, $u = L/W_0(L/e)$, $D = \log u$, and $c = \frac{1}{2} \log(2\pi) - \log 4$, the explicit center is

$$v_*(X) = u - \frac{1}{2} - \frac{c}{D} + \frac{97/24 - c^2/(2D^2)}{uD}. \quad (1.6)$$

For some constant $K > 0$ and all sufficiently large X ,

$$\left\lfloor v_*(X) - \frac{K}{u^2 D} \right\rfloor + 1 \leq N(X) \leq \left\lceil v_*(X) + \frac{K}{u^2 D} \right\rceil. \quad (1.7)$$

The constants and onset are not made effective here, and the floor and ceiling cannot be replaced uniformly by a single unqualified ceiling of a truncated expression.

Why the proof is not circular. The series A occurs on the right of (1.2), but only as a formal device for collecting stabilized coefficients. For each fixed R we first prove an approximation involving the known simple-permutation series and two finite polynomials. Only those series are submitted to the analytic asymptotic-transfer theorem. The resulting coefficients stabilize as the finite cutoff grows. No closure theorem is applied to A before its factorial expansion has been established.

1.1 Provenance, sources and reading conventions

[write] **Added 5 October 2026, batch 106.** *Provenance.* This report is Report 153 of the session bundle of Reports 1–243 that ProveIt’s docs/incoming intake processed as batch 106: the archive `Permutation_Graph_All_Orders_Expansion_Inverses_and_Sampling_Source.zip` (790,477 bytes, 23 files at the archive root, no wrapper directory), arrival commit 60f54ea06, placed in 47fc7a069. The text is the delivered `Report153.tex` (1413 lines), shipped as `article.tex`. The delivered 22-page PDF and the checksum manifest SHA256SUMS are not shipped (all 22 manifest entries verify); both are retrievable from the arrival commit. The package names no ProveIt commit or path, so the report has no pin. It names no person, tool or addressee and carries no “prepared for private review” line: the phrase “private review” occurs only in packaging negations such as “It contains neither unrelated earlier reports nor private review documents” (Section 11). It is a single-source report, so no merge choice was made. The write prefixed the 72 delivered labels with `prg:` and updated every reference; it added this subsection, the notes marked [write], Remark 8.2, Section 12.1, two macros and one bibliography entry. No statement, proof or number of the source was changed: the added remark is the last numbered statement of its section and the added displays are unnumbered, so every theorem and equation keeps its delivered number.

The sources, as the write read them (5 October 2026).

- OEIS A123448 [8]: revision 35 of 4 July 2026, now named “Number of permutation perfect graphs on n nodes” (the bibliography below keeps the name the source inspected); 14 terms, through $a_{14} = 14,974,215,412$. Its extensions credit a_{11} to Jakub Jabłoński (7 September 2020) and a_{12} – a_{14} to Tom Johnston (25 October 2020).
- Johnston [9]: the blog post “Enumerating permutation graphs”, dated 25 October 2020, at the address given in the bibliography. It reports a computer enumeration through $n = 14$ (75,912,033, 1,029,974,969 and 14,974,215,412 for $n = 12, 13, 14$, the OEIS values) and says that its author did not know whether the count is $o(n!)$. See the note in Section 10.
- Bassino, Bouvel, Féray, Gerin and Pierrot, arXiv:2402.06394v1, dated 9 February 2024 [5]: Proposition 4.6 (a modular-prime permutation graph has at most four realizers, all simple) and the proof of their Theorem 2.1 in Section 4.3, which proves $\#\{\text{permutation graphs on } n \text{ vertices}\} \geq n!/30$ for all large n from Proposition 4.6 and the Albert–Atkinson–Klazar asymptotic $s_n \sim e^{-2}n!$. These are the “coarse bounds” of Section 10.
- The repository’s transseries volume [10]: `p0:prop:factorial-core`, `p0:thm:core-reversion`, `p0:def:three-inverses` and `p0:thm:staircase`, used in Remark 8.2.
- Not read by the write: Borinsky [3], Albert–Atkinson–Klazar [1], Habib–Paul [4] and the 1990 paper [2], which the source read (the last including its Table 3); El-Zahar–Sauer [6] and Winkler [7], of which the source read only the publisher abstracts; the final journal text of [5], which the source could not obtain.

What was checked at intake. The batch-106 dossier read the manuscript in full and found no error. It recomputed $k!h_k$ for $k = 2, \dots, 7$, the expansion $U(y)^{-1} - 1 = 4y + 17y^2 + O(y^3)$, the second total-variation coefficient -15 and the residual $-97/24$, and compared the expansion with the OEIS terms and the six table values quoted in Section 10. On copies, the default companion replay and the full replay (both enumerators, all inputs through order nine; 4 min 27 s) reproduced the delivered result files byte for byte. At the write, an independent exact program recomputed, from the printed inputs alone, the series Q , $h_0, \dots, h_7$, the inverse-power coefficients of U through degree seven, $g_0, \dots, g_6$ and $z_1, \dots, z_7$: all agree with the source. It also obtained h_8 (note at the end of Section 7).

Relation to the repository. Before batch 106 no file outside the incoming archives treated A123448 or the enumeration of permutation graphs, and the source's only remark about the repository was then true (see the note in Section 10). The report meets the following texts.

- Two reports placed in the same commit and written separately in batch 106 are *siblings sharing a pipeline*: **a156808-circle-graphs** (bundle Reports 150 and 152, unlabeled circle graphs, OEIS A156808 and A156809) and **a005975-interval-graphs** (bundle Report 133, unlabeled interval graphs, OEIS A005975 and A005976). All three count a representation model (permutations here; indexed matchings; interval orders), show that almost every graph has a recoverable prime core with bounded decorations whose representations form one full orbit of a symmetry group (the Klein four-group of Section 2.2 here; the dihedral group of order $4n$; a duality of order two), make the symmetric cores negligible, and invert with Lambert- W brackets that keep the rounding. They share no statement and no manuscript cites another; the decomposition theories (modular decomposition here and for interval graphs, split decomposition for circle graphs), the external inputs and the normalizations ($n!/4$, $(2n-1)!!/(4n)$, half the Fishburn numbers) differ. “Prime” means modular-prime here and split-prime in the circle-graph report.
- **a007716-bipartite-multigraphs** and **a307316-leafless-multigraphs** use the same rare-symmetry philosophy for multigraphs and share no statement with this report.
- The transseries volume **Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/(transseries_and_inversion.tex)**: the Lambert- W centre of Section 8 is an instance of its factorial core, and its two shifts are the first coefficients of its formal core reversion (Remark 8.2, which also says what is not an instance).

No Lean or Rocq file of the repository treats permutation graphs or modular decomposition; no statement of this report is formalized, and its place in the collection confers no formal status.

[write] What is not claimed, collected from the source. Every truncation order is fixed: no bound on the growth of the constants with R , no growing-order truncation, no convergence or Borel summability of H , no identification of exponentially small sectors, no optimal truncation (Remark 6.2). No effective remainder constant, onset or certified threshold; the floor and ceiling of (1.7) and (8.8) cannot be replaced uniformly by one ceiling. Total variation controls events and uniformly bounded observables only, additively. The finite computations certify inputs and rational identities, not the analytic proof. No first-ever claim for the leading equivalent; the exact recurrence and table of [2], the poset asymptotics of [6] and the sampling models of [7] are prior art or close antecedents; the final journal text of [5] was not inspected, so overlap with it is unresolved. The table values for $n = 15, \dots, 20$ are quoted prior art, not verified here (exact verification stops at order nine). The repository search was “a bounded absence check, not a universal novelty proof”. “No OEIS submission, external publication, or public repository modification is part of this report.” The write adds: its numerical comparisons (Section 10) are observations and prove nothing about the remainders or the table.

Reading conventions (letters the manuscript reuses, with the tempting false readings; no symbol was renamed).

Symbol	Meaning here (where)	Not to be read as
a_n, r_n	unlabeled permutation graphs (A123448); vertex-rooted classes, i.e. vertex orbits summed over classes	r_n as na_n
A, C, F, T, Q	the formal series (1.1); $T = F^{\langle -1 \rangle}$, $Q = T \circ A$	Q as a quotient graph; C as the polynomial C_R
H, h_k	(1.2): coefficients in the basis $(n - k)!$	the inverse-power coefficients of U ($-94/3$ against $-97/3$); the circle-graph sibling's $h_n = (2n - 1)!!/(4n)$
$S(z), s_m$	simple permutations, $m \geq 4$ (2.2)	symmetric matchings or self-dual orders (siblings)
prime, simple	modular-prime graph; permutation without nontrivial proper interval	split-prime (circle graphs)
$f(\cdot), t(L, v)$	numbers of realizers; marked realizers (9.2)	
B_R, b_R, b_j	the polynomial (5.1); $b_R = B_R(1)$ (proof of Lemma 5.1); one-realizer counts (9.4)	each other
C_R, c_j, c	the polynomial (5.1); marked contexts with $t = 1$; $c = \frac{1}{2} \log(2\pi) - \log 4$	each other
J_m, J, J_R	the symmetric-permutation bound (5.2); the one-realizer series (9.5) and its truncation	each other
$K, K(x), K_R$	the constant in (1.7); the marked-context series; K_R both its truncation (Section 9) and the error constant in the proof of Theorem 8.1	each other
$E(x), E_k, E_n$	$1/x - 1/Q$; the mean number of length- k intervals (Lemma 3.1); events (Section 9)	each other
$\mathcal{D}, D$	the asymptotic derivative of Section 2.3; $D = \log u$	each other
$U, \mathcal{U}_n, \mathcal{U}_R(n)$	the series (8.2); the uniform law on classes; the constructed subclass	each other
$\mathbf{P}_n, P, P(x)$	the law of $\text{Inv}(\pi)$; a prime graph; a prefactor (proof of Corollary 6.1)	each other
$g(v), g_k$	the smooth log-approximant (Section 8); four-realizer coefficients (9.6)	each other
$N(X), \nu_R, \mathcal{A}_R$	the threshold; the large root of $\mathcal{A}_R(v) = X$; the approximant (8.5)	$\mathcal{A}_R$ as an interpolation of a_n : $\mathcal{A}_R(n) \neq a_n$ in general
u, L, D, v_*	$L = \log X$, $u = L/W_0(L/e)$, $D = \log u$, the centre (1.6)	L as a graph (context (L, v))
R	a fixed truncation order	the coefficient ring R of [10]
$Z(y), z_k, Z_n$	the total-variation series (9.12); a bounded statistic	each other

2 Structural conventions and established inputs

2.1 Intervals and modular decomposition

An interval of a permutation is a consecutive set of positions whose values also form a consecutive set. Its length may be one or the whole permutation; a *nontrivial proper interval* has length between 2 and $n - 1$. A permutation of order at least four is *simple* when it has no nontrivial proper interval.

A module in a graph G is a nonempty vertex set M such that every vertex outside M is adjacent to all of M or to none of M . Two sets *overlap* when their intersection and both differences are nonempty. A module is *strong* when it overlaps no module. The strong modules form the canonical modular decomposition tree. At its internal nodes, maximal proper strong modules are the child blocks. The quotient is either edgeless (parallel), complete (series), or modular-prime. We use *prime* only in this modular sense, not in the sense of graph split decomposition.

Inflating a permutation replaces each of its entries by an interval carrying a smaller permutation. Its canonical substitution tree has simple nodes of arity at least four and increasing or decreasing nodes of arity at least two. Consecutive monotone nodes with the same sign are flattened. The standard decomposition and the simple-permutation generating equation are due to Albert–Atkinson–Klazar [1].

The precise bridge from permutations to graphs is important. Habib–Paul [4, Lemma 20, printed p. 55] identify the strong modules of an inversion graph with the strong common intervals of its two-order representation. Consequently the canonical substitution tree agrees with the graph’s modular decomposition tree: simple nodes become prime quotients, and monotone nodes become series or parallel nodes. This does *not* say that every graph module is an interval of every realizer. It says that strong modules are exactly the canonical interval blocks relevant here.

Permutation graphs are closed under induced subgraphs and graph substitution. The latter follows directly by inflating a permutation realizer of the quotient by realizers of the child graphs. In particular, every graph counted by a context-and-decoration construction below is still a permutation graph.

2.2 Four orientation pairs and weighted prime counts

A permutation graph is a comparability graph whose complement is also a comparability graph. For a prime permutation graph of order at least four, the graph has two transitive orientations, reverse to one another, and its complement likewise has two. Each of the four pairs determines two total orders and hence a normalized permutation realizer. These classical facts and their use in permutation-graph enumeration are reviewed in [2, 5].

Here a *normalized realizer* is an actual permutation in $\mathcal{S}_m$ whose inversion graph belongs to the specified unlabeled graph class. No external labeling or ordering of the abstract graph’s vertices is counted as additional data.

Lemma 2.1 (Weighted four-realizer identity). *If P is a prime permutation graph on $m \geq 4$ vertices, then*

$$f(P) := \#\{\pi \in \mathcal{S}_m : \text{Inv}(\pi) \simeq P\} = \frac{4}{|\text{Aut}(P)|}.$$

If s_m is the number of simple permutations of order m , then

$$\sum_{[P] \text{ prime}, |P|=m} \frac{1}{|\text{Aut}(P)|} = \frac{s_m}{4}. \quad (2.1)$$

Proof. Fix a vertex set for P . An automorphism preserving both chosen transitive orientations preserves their union, a total order on that vertex set, so it is the identity. The automorphism action on the four orientation pairs is therefore free. Two pairs give the same normalized permutation exactly when they differ by an automorphism. This proves the first formula. Simplicity of a permutation is equivalent to modular primality of its inversion graph at these orders, by the canonical decomposition. Summing $f(P)$ gives s_m , proving (2.1). $\square$

The four realizer operations form a Klein group generated by inversion $\pi \mapsto \pi^{-1}$ and reverse-complement $\pi \mapsto \rho\pi\rho$, where $\rho(i) = m + 1 - i$. They preserve the unlabeled inversion graph. Complementing the graph is not an additional equivalence and supplies no extra factor of two. Orders one, two, and three do not enter the prime series.

2.3 The factorial asymptotic transfer used here

Let

$$S(z) = \sum_{m \geq 4} s_m z^m = \frac{z - z^2}{1 + z} - T(z). \quad (2.2)$$

For the factorial class with parameters $\alpha = \beta = 1$, write $\mathcal{D}f(x) = \sum_{k \geq 0} d_k x^k$ when

$$[x^n]f(x) = \sum_{k=0}^{R-1} d_k (n - k)! + O_R((n - R)!) \quad \text{for every fixed } R.$$

The basis is $(n - k)! = \Gamma(n + 1 - k)$, rather than $\Gamma(n - k)$. Polynomials have asymptotic derivative zero.

Borinsky's composition and inversion theorems [3, Theorem 35 and equations (62)–(67)] imply

$$\mathcal{D}S(z) = T'(z) \frac{z}{T(z)} \exp\left(\frac{1}{z} - \frac{1}{T(z)}\right). \quad (2.3)$$

In particular $\mathcal{D}S(0) = e^{-2}$. For a polynomial B with $B(0) = 0$ and $B'(0) = 1$, the specialization needed here is

$$\mathcal{D}(S \circ B)(x) = \frac{x}{B(x)} \exp\left(\frac{1}{x} - \frac{1}{B(x)}\right) (\mathcal{D}S)(B(x)). \quad (2.4)$$

For a polynomial C_0 , multiplication gives

$$\mathcal{D}(C_0(S \circ B)) = C_0 \mathcal{D}(S \circ B). \quad (2.5)$$

The additional chain-rule term in the general theorem vanishes because $\mathcal{D}B = 0$. Equations (2.3)–(2.5) are established asymptotic results, not identities inferred solely from formal power-series manipulation. We state precisely these external inputs and prove the graph-specific finite approximation next.

3 Excluding middle intervals

Lemma 3.1 (Interval tail). *For every fixed integer $R \geq 1$,*

$$\sum_{k=R+2}^{n-R} \frac{(n - k + 1)^2}{\binom{n}{k}} = O_R(n^{-R}). \quad (3.1)$$

Consequently at most $O_R((n - R)!)$ permutations of order n have an interval of some length in $[R + 2, n - R]$.

Proof. For each interval of k positions, its image set is uniform over all $\binom{n}{k}$ subsets of size k . Exactly $n - k + 1$ of these subsets are intervals of values. There are $n - k + 1$ intervals of positions, so the expected number of length- k intervals is

$$E_k = \frac{(n - k + 1)^2}{\binom{n}{k}}.$$

For $k \leq n/4$,

$$\frac{E_{k+1}}{E_k} = \frac{(k+1)(n-k)}{(n-k+1)^2} \leq \frac{k+1}{n-k+1} \leq \frac{1}{2}$$

for all sufficiently large n . The lower-end tail is therefore bounded by a constant times $E_{R+2} = O_R(n^{-R})$. At the upper end put $\ell = n - k$ and $b_\ell = (\ell + 1)^2 / \binom{n}{\ell}$. Then

$$\frac{b_{\ell+1}}{b_\ell} = \frac{(\ell + 2)^2}{(\ell + 1)(n - \ell)}.$$

For $1 \leq \ell \leq n/4$ this is at most $1/2$ once n is large. The upper-end tail starts at $b_R = O_R(n^{-R})$. Throughout the remaining central band, $\binom{n}{k} \geq \binom{n}{\lfloor n/4 \rfloor}$, up to an immaterial shift of one in a boundary index. This grows exponentially in n , whereas the numerator and number of terms are polynomial. The central sum is exponentially small. This proves (3.1). A union bound and multiplication by $n!$ complete the proof, since $n!n^{-R} = O_R((n - R)!)$. $\square$

Definition 3.2. A permutation is R -good if it has no interval of a length in $[R + 2, n - R]$. All goodness statements below have R fixed and n sufficiently large in terms of R .

The argument is a probability estimate on uniform *permutations*. It is not an assertion that such permutations induce uniform graph classes. Later, an exceptional graph class will be charged to one of its exceptional realizers; this elementary surjection bound is all that is needed to transfer the error count to graphs.

4 Recovering one large prime node

Lemma 4.1 (Large simple node). *The canonical substitution tree of an R -good permutation has a unique internal node of arity greater than $R + 1$. That node is simple. Its subtree has at least $n - R + 1$ leaves, each of its children has at most $R + 1$ leaves, and at most $R - 1$ leaves lie outside it.*

Proof. Every proper canonical subtree is an interval, so its leaf count is at most $R + 1$ or at least $n - R + 1$. Starting at the root, follow a child of the latter size for as long as one exists. Such a child is unique, because two would have total size exceeding n . At the terminal node v , the subtree has $N \geq n - R + 1$ leaves and all children have at most $R + 1$ leaves.

The node cannot be monotone. Otherwise any consecutive union of its children is also a permutation interval. Accumulate children from one end until their total size first exceeds $R + 1$. The resulting interval has size between $R + 2$ and $2R + 2$. For $n > 3R + 2$, this lies in the forbidden band and is proper. Notice that the union need not be a canonical subtree: being a proper *permutation interval* is the property used in the contradiction.

Thus v is simple and its arity is at least $N/(R + 1)$, eventually greater than $R + 1$. Any descendant internal node lies inside a child with at most $R + 1$ leaves. There are at most $R - 1$ leaves outside v ,

so every ancestor has at most R children: one on the path to v and at most $R - 1$ off that path. Every off-path node is likewise bounded. This proves uniqueness and all size assertions. $\square$

To turn this into an unlabeled product, the large inserted object must be intrinsically recoverable in every bounded context.

Lemma 4.2 (Connected and coconnected insertion). *Suppose a graph H is substituted at one vertex of a graph K . If H and its complement are both connected, the set M of inserted vertices is a strong module of the resulting graph.*

Proof. It is a module by construction. Suppose a module N overlaps M . Let $X = M \cap N$ and choose $y \in N \setminus M$. Since M is a module, y has constant adjacency $\eta \in \{0, 1\}$ to M . For any $z \in M \setminus N$, the module property of N forces z to have the same adjacency to every $x \in X$ as it has to y . Thus every edge across the nontrivial cut $(X, M \setminus N)$ has value η . If $\eta = 0$, H is disconnected; if $\eta = 1$, its complement is disconnected. Both contradict the hypothesis. $\square$

A prime quotient of order at least four is connected and coconnected. Inflating all its vertices by nonempty graphs preserves both properties: paths in the quotient connect different blocks, and a vertex in another adjacent block connects vertices within a block if necessary. Apply the same argument in the complement. Thus Lemma 4.2 applies to every prime-rooted inflation used here, regardless of the surrounding context.

Proposition 4.3 (Unlabeled context recovery). *Fix R . Consider the following construction at order n :*

1. *choose a vertex-rooted permutation graph on $d + 1$ vertices, where $0 \leq d \leq R - 1$;*
2. *choose a prime permutation-graph quotient of order $m > R + 1$;*
3. *decorate each quotient vertex by a nonempty permutation graph of order at most $R + 1$, modulo quotient automorphisms;*
4. *insert the decorated quotient at the marked context vertex.*

For sufficiently large n , the output graph recovers exactly these data, up to the stated isomorphisms. Every graph realized by an R -good permutation occurs among the outputs.

Proof. The inserted vertex set is a strong module by Lemma 4.2. Within it, the maximal proper strong modules are the decorated ports and the quotient is the chosen prime graph. All decoration nodes have arity at most $R + 1$, as do all nodes of the context, whose order is at most R . Hence the chosen quotient is the unique node of arity greater than $R + 1$ in the output's canonical modular tree. It is intrinsically recognizable, including when the context consists only of its root.

An isomorphism between output graphs must preserve this unique node and its strong module. The induced graph on that module recovers the inner decorated quotient, with ports identified up to quotient automorphism. Collapsing it to one marked vertex recovers the context up to root-preserving isomorphism. Conversely each listed isomorphism of data gives an isomorphism of outputs. There is no extra division by a context root stabilizer and no choice of a transitive orientation at the context root.

Finally Lemma 4.1 and the Habib–Paul correspondence identify exactly such data inside every R -good realizer. Taking induced subgraphs and collapsing interval blocks preserves the permutation-graph property. Thus its graph lies among the outputs. $\square$

Let $\mathcal{U}_R(n)$ be this exact subclass of graph classes. It may contain graphs with some bad realizers. No converse goodness assertion is needed. The two inclusions used are simply

$$\mathcal{U}_R(n) \subseteq \{\text{all graph classes of order } n\}, \quad \text{Inv}(\{R\text{-good permutations}\}) \subseteq \mathcal{U}_R(n).$$

Every class outside $\mathcal{U}_R(n)$ therefore has only bad realizers. Selecting one realizer per such class and applying Lemma 3.1 gives

$$0 \leq a_n - |\mathcal{U}_R(n)| = O_R((n - R)!). \quad (4.1)$$

The rooted context factor is precisely r_{d+1} . Port decorations, by contrast, are *unrooted*, since outside adjacency is uniform throughout a port module.

5 Burnside weights and negligible symmetric quotients

Define finite polynomials

$$B_R(x) = \sum_{j=1}^{R+1} a_j x^j, \quad C_R(x) = \sum_{d=0}^{R-1} r_{d+1} x^d. \quad (5.1)$$

For a fixed prime graph P with m vertices, graph decorations of its ports are acted on by $\text{Aut}(P)$. Burnside's lemma says that their orbit generating series is the average of the fixed-decoration series. The identity contribution is $B_R(x)^m / |\text{Aut}(P)|$. Summation over prime classes using (2.1) gives exactly $S(B_R(x))/4$ for these identity contributions.

The identity contribution is a rational weight, not in general an integer count of a graph family. We must bound the missing nonidentity Burnside contributions before using it in an asymptotic formula.

Lemma 5.1 (Symmetry bound). *The number of permutations of order m fixed by at least one nonidentity Klein-group operation is at most*

$$J_m = \exp\left(\frac{m}{2} \log m + O(m)\right). \quad (5.2)$$

For fixed R , all decorated symmetric prime-quotient contributions at total graph order n , including bounded contexts, have total size at most

$$\exp\left(\frac{n}{2} \log n + O_R(n)\right) = o((n - R)!). \quad (5.3)$$

Proof. Inversion-fixed permutations are involutions. The product of inversion and reverse-complement is fixed when $\pi = \rho\pi^{-1}\rho$, equivalently $(\rho\pi)^2 = 1$, so it has the same count. Reverse-complement-fixed permutations commute with ρ . There are exactly $2^{\lfloor m/2 \rfloor} \lfloor m/2 \rfloor!$ of them: permute the reversal pairs and independently choose their directions, fixing the middle point at odd order.

If I_m counts involutions, its exponential generating function is $\exp(z + z^2/2)$. Positivity of its coefficients gives, at $t = \sqrt{m}$,

$$I_m \leq m! \frac{\exp(t + t^2/2)}{t^m} = \exp\left(\frac{m}{2} \log m + O(m)\right).$$

The reverse-complement count satisfies the same bound by Stirling's formula. A union bound proves (5.2).

If P has a nontrivial automorphism, Lemma 2.1 shows that it has fewer than four distinct realizers. Its Klein orbit therefore has a nontrivial stabilizer. Such prime classes can be injected into the union just counted by choosing one of their realizers.

Put $b_R = \sum_{j=1}^{R+1} a_j$. The number of decoration assignments at m distinguishable ports, ignoring total size, is at most b_R^m . This also bounds each fixed-assignment term, and $|\text{Aut}(P)| \leq 4$. Summing over $m \leq n$ and multiplying by the bounded number $\sum_{d < R} r_{d+1}$ of contexts gives the first bound in (5.3). Since $\log((n-R)!) = n \log n - n + O_R(\log n)$, the second assertion follows. $\square$

Theorem 5.2 (Finite-polynomial approximation). *For every fixed $R \geq 1$,*

$$a_n = \frac{1}{4} [x^n] C_R(x) S(B_R(x)) + O_R((n-R)!). \quad (5.4)$$

Proof. Proposition 4.3 makes the class construction an ordinary unlabeled product of contexts and decorated quotients. Burnside's identity terms give the displayed coefficient. Lemma 5.1 bounds all nonidentity terms by an error smaller than every fixed factorial order. The series S also contains quotients with arity $m \leq R+1$; these contribute only orders at most $(R+1)^2 + R - 1$ after decoration and context insertion, so they disappear at sufficiently large n . Finally use (4.1). $\square$

The theorem has no hidden hypothesis about the unknown growth of a_n or r_n : for fixed R , they enter only as finitely many integer coefficients of B_R and C_R . The constants may depend on those finite inputs and on R .

6 Factorial transfer and all orders

We now prove Theorem 1.1. Apply (2.4) and (2.5) to Theorem 5.2. For a fixed R , the coefficients through degree $R-1$ are generated by

$$H_R(x) = C_R(x) \frac{x}{B_R(x)} \exp\left(\frac{1}{x} - \frac{1}{B_R(x)}\right) (\mathcal{D}S)(B_R(x)). \quad (6.1)$$

Precisely,

$$a_n = \frac{1}{4} \sum_{k=0}^{R-1} [x^k] H_R(x) (n-k)! + O_R((n-R)!).$$

The finite graph approximation and the factorial-transfer remainder have the same order and can be combined.

Because $A - B_R = O(x^{R+2})$ and both have leading coefficient one,

$$\frac{1}{A} - \frac{1}{B_R} = O(x^R), \quad \frac{x}{A} - \frac{x}{B_R} = O(x^{R+1}).$$

Also $C - C_R = O(x^R)$, and ordinary formal substitution into $\mathcal{D}S$ changes no coefficient below degree R . Hence

$$H_R(x) \equiv C(x) \frac{x}{A(x)} \exp\left(\frac{1}{x} - \frac{1}{A(x)}\right) (\mathcal{D}S)(A(x)) \pmod{x^R}. \quad (6.2)$$

Substitute (2.3). The exponentials combine by cancellation of $1/A$, and $T'(A) = Q'/A'$ by the formal chain rule. Thus the right side of (6.2) is exactly (1.2). This proves (1.3) at every fixed order.

For well-definedness, $a_1 = 1$ and $a_2 = 2$, while $F(x) = x + 2x^2 + \dots$. Solving $F(Q) = A$ gives $Q = x + O(x^3)$. Therefore

$$E(x) := \frac{1}{x} - \frac{1}{Q(x)} \in x\mathbb{Q}[[x]], \quad \frac{xQ'}{QA'} \in 1 + x\mathbb{Q}[[x]],$$

and $C(0) = r_1 = 1$. In particular the constant terms in the simple-permutation transfer, e^{-2} from $\mathcal{DS}$ and e^2 from $B_R = x + 2x^2 + \dots$, cancel. This explains why the leading graph constant is $1/4$ rather than $e^{-2}/4$.

The degree- k exponent coefficient uses Q only through degree $k + 2$. Since $F(Q) = A$ has a unit linear term, these coefficients use only $a_1, \dots, a_{k+2}$. The prefactor needs no more data, and its C factor uses $r_1, \dots, r_{k+1}$. This proves the finite dependency assertion.

Corollary 6.1 (Denominators). *Every coefficient in (1.2) satisfies $k!h_k \in \mathbb{Z}$.*

Proof. The inverse of a series $x + O(x^2)$ with integer coefficients has integer coefficients: solving successive coefficients involves no division other than by its linear coefficient one. Thus T , Q , and Q/x have integer coefficients. The units Q/x and A' have constant term one, so

$$P(x) := C(x) \frac{xQ'(x)}{Q(x)A'(x)} \in \mathbb{Z}[[x]].$$

Moreover $Q = x + O(x^3)$ implies $E \in x\mathbb{Z}[[x]]$. In degree k , $\exp E = \sum_{j=0}^k E^j/j!$ has denominator dividing $k!$. Multiplication by P preserves this property, because $j!$ divides $k!$ for $j \leq k$. $\square$

Remark 6.2 (The meaning of all orders). The quantifiers are: for each fixed R there exist a constant and an onset depending on R . The proof does not bound their growth with R and does not justify substituting $R = R(n)$, choosing a least term, claiming convergence or Borel summability of H , identifying all exponentially small sectors, or estimating an exponentially small remainder by optimally truncating the series. Finite exact coefficient algorithms and an all-orders asymptotic statement do not establish any of those stronger conclusions.

7 Exact coefficients and finite certification

The necessary small graph data are as follows. The unrooted and rooted columns were obtained by two separately implemented exhaustive canonicalization routes; no source-table value beyond these orders is used in the asymptotic calculation.

n	a_n	r_n	s_n	Prime graph classes
1	1	1	0	0
2	2	2	0	0
3	4	6	0	0
4	11	20	2	1
5	33	89	6	3
6	142	504	46	16
7	776	3776	338	92
8	5699	34620	2926	772
9	50723	368683	28146	7122

The zero values in the prime columns below order four reflect the convention that the prime series starts at four. Both enumerations check the unrooted and vertex-rooted columns; the independent refined-color route additionally supplies the displayed simple and prime counts. In particular all inputs for $h_0, \dots, h_7$, namely $a_1, \dots, a_9$ and $r_1, \dots, r_8$, have two exhaustive replays. The extra r_9 does not by itself determine h_8 , which would also require a_{10} .

Solving $F(Q) = A$ gives

$$Q(x) = x - 2x^3 - 5x^4 - 31x^5 - 180x^6 - 1418x^7 - 12327x^8 - 121659x^9 + O(x^{10}).$$

The coefficient table is

k	h_k	$k!h_k$
0	1	1
1	-4	-4
2	-1	-2
3	-94/3	-188
4	-769/6	-3076
5	-19969/15	-159752
6	-531812/45	-8508992
7	-40114096/315	-641825536

For clarity about the first correction, one can calculate directly:

$$Q = x - 2x^3 + O(x^4), \quad E = -2x + O(x^2), \quad C \frac{xQ'}{QA'} = 1 - 2x + O(x^2).$$

Hence $H = 1 - 4x + O(x^2)$. Both the rooted context and the prime inflation contribute; replacing C by an unrooted series would produce incorrect higher coefficients.

7.1 Two exact formal calculation routes

One route computes Q recursively from $F(Q) = A$ and evaluates (1.2) with rational truncated-series arithmetic. A second route obtains T by Lagrange inversion,

$$[z^m]T(z) = \frac{1}{m}[w^{m-1}] \left(\frac{w}{F(w)} \right)^m, \quad (7.1)$$

then evaluates (6.1) for successive finite cutoffs. The factor e^{-2} in (2.3) can be carried symbolically and canceled with the constant e^2 in the composition factor. Every remaining calculation is rational. The routes agree through h_7 , and the finite-cutoff route checks the stabilization in (6.2) for $R = 1, \dots, 8$.

7.2 Two graph enumeration routes

Each route enumerates all $n!$ permutations, constructs its inversion graph, and computes an exact isomorphism key. The first partitions vertices by degree (and, for rooted keys, root status), then minimizes an adjacency encoding over all permutations within the cells. For each unrooted representative it counts distinct rooted keys. The second refines invariant color classes before exhaustively minimizing within them; all minimizing labelings recover the vertex orbits and automorphism count. Both are exhaustive rather than randomized isomorphism heuristics.

Additional finite checks verify the weighted identity $|\text{Aut}(P)|f(P) = 4$ and equality of the realizer set with the Klein orbit for every prime graph through order nine. Direct subset-module checks at smaller orders confirm the prime/simple correspondence. These computations are useful normalization tests. They do not replace the structural theorems or certify an asymptotic remainder constant.

[write] **The coefficient h_8 , 5 October 2026.** By Theorem 1.1, h_8 depends only on $a_1, \dots, a_{10}$ and $r_1, \dots, r_9$. The table above supplies all of them except a_{10} , and OEIS A123448 lists $a_{10} = 524,572$, a value that the companion does not enumerate. With it, the coefficient of x^{10} in Q is $-1,319,072$, and evaluating (1.2) in exact rational arithmetic gives

$$h_8 = -\frac{3687969659}{2520}, \quad 8!h_8 = -59,007,514,544,$$

an integer, as Corollary 6.1 requires. The same program reproduces $h_0, \dots, h_7$ from the table. The value of h_8 depends on the OEIS term a_{10} being correct; h_9 would also need r_{10} , which is not known here. (Added after the independent check of 5 October 2026: the value $a_{10} = 524,572$ was confirmed by an exhaustive canonical enumeration of the inversion graphs of all $10!$ permutations, which also reproduced $a_1, \dots, a_9$ and $r_1, \dots, r_9$ of the table, and an independent exact program recomputed h_8 and $8!h_8$ from these values. The caveat of the previous sentence is therefore discharged.)

8 Threshold inversion with rounding preserved

8.1 Strict increase and ordinary inverse powers

Adding an isolated vertex gives an injection from graph classes of order n to those of order $n+1$: remove one isolated vertex to recover the original class. The complete graph on $n+1$ vertices is outside the image for $n \geq 1$. Thus

$$a_{n+1} > a_n \quad (n \geq 1). \quad (8.1)$$

The sequence tends to infinity by Theorem 1.1, so $N(X)$ is well defined for all real $X > 0$.

Let $(n)_k = n(n-1) \cdots (n-k+1)$, with $(n)_0 = 1$, and define a formal inverse-power series

$$U(y) = \sum_{k \geq 0} h_k \frac{y^k}{\prod_{j=0}^{k-1} (1 - jy)}. \quad (8.2)$$

At each degree this is a finite sum. Its beginning is

$$U(y) = 1 - 4y - y^2 - \frac{97}{3}y^3 - \frac{1339}{6}y^4 - \frac{11603}{5}y^5 + O(y^6). \quad (8.3)$$

Using a formal logarithm and Stirling's expansion gives, in the usual fixed-order asymptotic sense,

$$\begin{aligned} \log a_n &= n(\log n - 1) + \frac{1}{2} \log n + c - \frac{47}{12n} - \frac{9}{n^2} + O(n^{-3}), \\ c &= \frac{1}{2} \log(2\pi) - \log 4. \end{aligned} \quad (8.4)$$

All further inverse-power coefficients follow from (8.2), $\log U$, and the Bernoulli-number terms in Stirling's formula. For example $-47/12 = 1/12 - 4$ and $-9 = -1 - (-4)^2/2$.

8.2 An arbitrary fixed-order smooth approximation

For a fixed $R \geq 1$, set

$$\mathcal{A}_R(v) = \frac{\Gamma(v+1)}{4} \sum_{k=0}^{R-1} \frac{h_k}{(v)_k}, \quad (v)_k = \prod_{j=0}^{k-1} (v-j). \quad (8.5)$$

For sufficiently large real v , the finite factor is positive and

$$(\log \mathcal{A}_R)'(v) = \log v + O_R(v^{-1}) > 0. \quad (8.6)$$

For all sufficiently large X , let $\nu_R(X)$ be the unique large solution of $\mathcal{A}_R(v) = X$. The qualifier “large” excludes irrelevant behavior of the rational factor near its poles at small integers. The factorial expansion implies

$$\log a_n - \log \mathcal{A}_R(n) = O_R(n^{-R}). \quad (8.7)$$

Theorem 8.1 (Threshold window). *For each fixed $R \geq 1$, there is a positive function*

$$\varepsilon_R(X) = O_R \left(\frac{\nu_R(X)^{-R}}{\log \nu_R(X)} \right)$$

such that, for all sufficiently large X ,

$$\lfloor \nu_R(X) - \varepsilon_R(X) \rfloor + 1 \leq N(X) \leq \lceil \nu_R(X) + \varepsilon_R(X) \rceil. \quad (8.8)$$

In particular $N(X) = \lceil \nu_R(X) \rceil$ whenever $\nu_R(X)$ stays farther than $\varepsilon_R(X)$ from every integer.

Proof. Choose K_R and n_R so that the absolute error in (8.7) is at most $K_R n^{-R}$ for integers $n \geq n_R$. Put $v = \nu_R(X)$. On $[v/2, 2v]$, equation (8.6) gives $(\log \mathcal{A}_R)' \geq \frac{1}{2} \log v$ once v is large. Choose

$$\varepsilon_R = D_R \frac{v^{-R}}{\log v}, \quad D_R > 2^{R+1} K_R.$$

If an integer n in that interval satisfies $n \leq v - \varepsilon_R$, then

$$\log \mathcal{A}_R(n) - \log X \leq -\frac{1}{2} \varepsilon_R \log v < -2^R K_R v^{-R} \leq -K_R n^{-R}.$$

Together with (8.7), this yields $a_n < X$. The same argument with the opposite sign gives $a_n > X$ for integers $n \geq v + \varepsilon_R$ in the interval. Monotonicity (8.1) extends these comparisons to integers outside it, using nearby integers still inside the interval. Thus every integer at or below the lower edge is too small, and every integer at or above the upper edge is sufficient. Taking floor and ceiling yields (8.8), including integer-edge cases because the comparison inequalities are strict. $\square$

8.3 The explicit Lambert W center

Let W_0 be the principal real Lambert function, so $W_0(z)e^{W_0(z)} = z$. With

$$L = \log X, \quad u = \frac{L}{W_0(L/e)}, \quad D = \log u, \quad (8.9)$$

one has $u(\log u - 1) = L$. Set

$$g(v) = v(\log v - 1) + \frac{1}{2} \log v + c - \frac{47}{12v}.$$

Equation (8.4) says $\log a_n = g(n) + O(n^{-2})$. Write $\delta_0 = -\frac{1}{2} - c/D$. The terms of order one in $g(u + \delta_0) - L$ cancel because $D\delta_0 + D/2 + c = 0$. Taylor expansion gives the remaining first inverse-power residual

$$g(u + \delta_0) - L = \frac{\delta_0^2/2 + \delta_0/2 - 47/12}{u} + O(u^{-2}) = \frac{-97/24 + c^2/(2D^2)}{u} + O(u^{-2}). \quad (8.10)$$

Canceling this with a second shift divided by the leading slope D gives the v_* in (1.6). Since $g'(v) = \log v + O(v^{-1})$, its residual is $O(u^{-2})$ and its smooth inverse error is $O(u^{-2}/D)$.

The same strict-comparison argument used for Theorem 8.1, now using $\log a_n - g(n) = O(n^{-2})$, proves (1.7). In particular the less refined center

$$u - \frac{1}{2} - \frac{c}{D}$$

has inverse error $O(1/(uD))$ before rounding. Under the alternate normalization $L_4 = \log(4X)$ and $t = L_4/W_0(L_4/e)$, its first shift is often written

$$t - \frac{\log(2\pi t)}{2 \log t}.$$

These are compatible after re-expansion; one must not mix $\log X$ and $\log(4X)$ while retaining the same constant c .

8.4 What the inverse does and does not certify

The coefficients and formal shifts can be computed to any fixed order. Once R is chosen, a large root of (8.5) can be approximated numerically, for example by Newton iteration initialized at a Lambert- W center. This is a method for evaluating the smooth approximation, not a certified algorithm for the exact integer threshold at an arbitrary supplied X : no explicit K_R or uniformly effective onset has been obtained here.

A staircase with jumps of one cannot be approximated uniformly by a continuous expression to $o(1)$. Near an integer center, the two neighboring integers in the bracket are genuine possibilities. Away from the shrinking boundary window the ceiling is stable. Neither additional formal coefficients nor higher floating-point precision remove this logical distinction.

Remark 8.2 ([write] Section 8 in the terms of the transseries volume, 5 October 2026). Statement by statement, against the repository's transseries volume [10]:

(a) *The threshold is the volume's integer staircase, and `p0:thm:staircase (5)` applies to it.* By (8.1) the sequence $(a_n)_{n \geq 1}$ is strictly increasing and tends to infinity, so $N(X)$ is the staircase N_* of

`p0: def: three-inverses` (1) with $n_0 = n_1 = 1$. Part (5) of `p0: thm: staircase` uses nothing else: for every Y and every continuous g on $[Y, \infty)$, $\sup_{X \geq Y} |N(X) - g(X)| \geq \frac{1}{2}$. This is the quantitative form of the statement in Section 8 that a staircase with jumps of one cannot be approximated uniformly by a continuous expression to $o(1)$.

(b) *Theorem 8.1 and the bracket (1.7) are not instances of parts (1)–(2) of `p0: thm: staircase`.* Those parts concern the inverse of an admissible interpolation, a continuous strictly increasing $\mathcal{A}$ with $\mathcal{A}(n) = a_n$ at every integer n . The approximant (8.5) satisfies $\mathcal{A}_R(n) = \frac{1}{4} \sum_{k < R} h_k(n - k)!$, because $\Gamma(n + 1)/(n)_k = (n - k)!$; this differs from a_n by $O_R((n - R)!)$ and is not equal to it in general, and the same holds for the centre v_* . The source brackets $N(X)$ directly at the integers, from (8.7) (or $\log a_n - g(n) = O(n^{-2})$) and the monotonicity (8.1). The result has the shape of the separation condition `p0: thm: staircase` (2), with $x_* = \nu_R(X)$ and $\eta = \varepsilon_R(X)$, but it is proved without an admissible interpolation and is not an instance of it.

(c) *The centre u is an instance of `p0: prop: factorial-core`.* With $\kappa = 1$, $d = -1$ and target $L = \log X > 0$, that proposition gives $v = W_0(L/e)$, the core solution $\exp(v + 1)$ of $X'(\log X' - 1) = L$ (unique under $\log X' - 1 > 0$) and the core slope $\kappa(v + 1) = \log \exp(v + 1)$. Since $W_0(L/e) \exp(W_0(L/e)) = L/e$, one has $\exp(v + 1) = L/W_0(L/e) = u$. So u is that core solution, as (8.9) states, and its core slope is $\log u = D$, the leading slope of Section 8. The factorial phase $X'(\log X' - 1)$ lies outside the volume’s exponential–power model, so `p0: thm: lambert-core` and `p0: thm: lambert-centered` do not apply.

(d) *The two shifts in v_* are the first two coefficients of the formal reversion `p0: thm: core-reversion`, after the change of variable $x = u(1 + E)$, $t = 1/u$.* Write the expansion (8.4) as $\log a_n \approx n(\log n - 1) + \frac{1}{2} \log n + c + \sum_{j \geq 1} q_j n^{-j}$, with $q_1 = -47/12$ and $q_2 = -9$. Since $u(D - 1) = L$,

$$x(\log x - 1) - L = u(DE + (1 + E) \log(1 + E) - E),$$

so the formal equation “ $\log a_x = L$ ” divided by u reads $\Lambda E + h(E) + f(t, E) = 0$ with

$$\Lambda = D, \quad h(E) = (1 + E) \log(1 + E) - E, \quad f(t, E) = t \left(\frac{D}{2} + c + \frac{1}{2} \log(1 + E) \right) + \sum_{j \geq 1} q_j t^{j+1} (1 + E)^{-j}.$$

Here $h \in E^2 R[[E]]$ and $f \in tR[[t, E]]$ (the volume’s variable u is E here) over $R = \mathbb{Q}[c, D, D^{-1}]$, with c and D treated as independent indeterminates, as the volume’s Remark `p0: rem: core-instances` treats the slope of its double factorial chapter. The triangular recurrence of `p0: thm: core-reversion` gives

$$e_1 = -\frac{1}{D} \left(\frac{D}{2} + c \right) = -\frac{1}{2} - \frac{c}{D}, \quad e_2 = -\frac{1}{D} \left(\frac{e_1^2}{2} + \frac{e_1}{2} + q_1 \right) = \frac{1}{D} \left(\frac{97}{24} - \frac{c^2}{2D^2} \right),$$

because $e_1(e_1 + 1)/2 = \frac{1}{2}(-\frac{1}{2} - \frac{c}{D})(\frac{1}{2} - \frac{c}{D}) = \frac{c^2}{2D^2} - \frac{1}{8}$ and $-\frac{1}{8} - \frac{47}{12} = -\frac{97}{24}$. Then $u(1 + e_1 t + e_2 t^2) = u + e_1 + e_2/u$ is the centre v_* of (1.6). This uses the theorem’s existence, uniqueness and triangular recurrence; its Bell form does not apply, since f contains $\log(1 + E)$, which is not of the form $\sum_k \phi_k(t)(1 + E)^{-\mu_k}$. The instance is formal only. The error $O(u^{-2}/D)$ of the smooth inverse and the bracket (1.7) rest on the source’s own Taylor and comparison arguments, which the volume’s analytic theorems, stated for the exponential–power model, do not cover.

9 All orders comparison of the sampling laws

Let $\mathcal{G}_n$ be the finite set of unlabeled permutation graphs of order n , and write $f(G)$ for the number of normalized permutation realizers of G . Then

$$P_n(G) = \frac{f(G)}{n!}, \quad U_n(G) = \frac{1}{a_n}, \quad \sum_{G \in \mathcal{G}_n} f(G) = n!. \quad (9.1)$$

Put $M_{4,n} = \#\{G \in \mathcal{G}_n : f(G) = 4\}$. We first derive an all-orders expansion for this exact fiber-size class, then transfer it to total variation.

9.1 Marked context realizers and exact multiplicativity

For a rooted graph (L, v) , define its marked-realizer count by

$$t(L, v) = \#\{(\kappa, i) : (\text{Inv}(\kappa), i) \simeq (L, v)\} = f(L) |\text{Aut}(L) \cdot v|. \quad (9.2)$$

For each unmarked realizer κ , the allowed marked positions are exactly the images of the automorphism orbit of v . This proves the second equality. Root positions are counted separately even when the underlying permutation is unchanged.

Lemma 9.1 (Exact fiber product). *Consider a graph $G \in \mathcal{U}_R(n)$ with asymmetric prime quotient P , port graphs $(D_p)_{p \in V(P)}$, and outer rooted context (L, v) . Then*

$$f(G) = 4t(L, v) \prod_{p \in V(P)} f(D_p). \quad (9.3)$$

Proof. Let $H = P[D_p : p \in V(P)]$ be the inserted inner graph. Its child modules are its maximal proper strong modules. In every realizer of H they are canonical common-interval blocks. Collapsing them gives one of the four distinct normalized simple realizers of P . For each one the identification with the ports of P is unique, because P has no automorphisms. Standardizing each interval gives an independent choice of a realizer of D_p . Conversely these choices reconstruct the permutation by inflation. Collapse and standardization recover the inputs, so $f(H) = 4 \prod_p f(D_p)$.

In G , the vertices of H form the distinguished strong module having the uniquely large prime quotient. This follows from Lemma 4.2 even when the context root belongs to a series or parallel node. Every realizer of G therefore identifies that module as a strong common interval. Collapsing it, retaining its position, and standardizing its interior gives a marked context realizer counted by $t(L, v)$ and an inner realizer counted by $f(H)$. Conversely their marked inflation reconstructs the realizer of G . The intrinsic recovery proves this is a bijection. $\square$

Every factor after the initial four in (9.3) is a positive integer. An asymmetric-core output has exactly four realizers if and only if every port graph has one realizer and its context has one marked realizer. The latter condition need not force a one-vertex context.

Define finite counts

$$\begin{aligned} b_j &= \#\{G \in \mathcal{G}_j : f(G) = 1\}, \\ c_j &= \#\{(L, v) : |L| = j, t(L, v) = 1\}, \end{aligned} \quad (9.4)$$

and formal series

$$J(x) = \sum_{j \geq 1} b_j x^j, \quad K(x) = \sum_{d \geq 0} c_{d+1} x^d. \quad (9.5)$$

Let J_R and K_R truncate these series at degrees $R + 1$ and $R - 1$, respectively.

Theorem 9.2 (Four-realizer factorial expansion). *Set $Q_J = T \circ J$ and*

$$H_4(x) = K(x) \frac{x Q'_J(x)}{Q_J(x) J'(x)} \exp\left(\frac{1}{x} - \frac{1}{Q_J(x)}\right) = \sum_{k \geq 0} g_k x^k. \quad (9.6)$$

For every fixed $R \geq 1$,

$$M_{4,n} = \frac{1}{4} \sum_{k=0}^{R-1} g_k (n-k)! + O_R((n-R)!). \quad (9.7)$$

The coefficient g_k uses only $b_1, \dots, b_{k+2}$ and $c_1, \dots, c_{k+1}$, and $k!g_k \in \mathbb{Z}$.

Proof. The same finite-core bounds as in Theorem 5.2, combined with Lemma 9.1, give

$$M_{4,n} = \frac{1}{4} [x^n] K_R(x) S(J_R(x)) + O_R((n-R)!). \quad (9.8)$$

Indeed graph classes outside $\mathcal{U}_R(n)$ number $O_R((n-R)!)$, hence bound the excluded four-realizer classes too. Symmetric large quotients contribute only $\exp((n/2) \log n + O_R(n))$ classes; this bounds both any genuine four-realizer contributions from them and their identity terms in the displayed approximation. At asymmetric quotients there are exactly four simple realizers and no port identifications; the fiber product selects precisely the port and context counts in J_R and K_R . Bounded quotients disappear at large n as before.

Apply the same polynomial composition theorem to (9.8) and stabilize the coefficients, with J, K replacing A, C . Since $b_1 = 1$, $b_2 = 2$, and $c_1 = 1$, one has $Q_J = x + O(x^3)$ and $H_4(0) = 1$. The finite-dependency and denominator arguments of Section 6 apply verbatim. No asymptotic property of J or K was assumed before finite transfer. $\square$

9.2 The second coefficient and exact finite inputs

At order three, the empty and complete graphs have one realizer each; the edge-plus-isolated-vertex graph and its complement have two each. Thus $(b_1, b_2, b_3) = (1, 2, 2)$. Both rooted contexts of order two have a root orbit of size two, giving $c_1 = 1, c_2 = 0$. At $R = 2$, (9.8) reduces to

$$M_{4,n} = \frac{1}{4} [x^n] S(x + 2x^2 + 2x^3) + O((n-2)!).$$

In the composition transfer the three factors begin

$$\begin{aligned} x/J_2(x) &= 1 - 2x + O(x^2), \\ \exp(1/x - 1/J_2(x)) &= e^2(1 - 2x + O(x^2)), \\ (\mathcal{DS})(J_2(x)) &= e^{-2}(1 - 4x + O(x^2)). \end{aligned}$$

Hence $g_1 = -8$ and

$$M_{4,n} = \frac{n!}{4} \left(1 - \frac{8}{n} + O(n^{-2})\right). \quad (9.9)$$

Full fiber enumeration and explicit vertex-orbit checks give

j	1	2	3	4	5	6	7	8
b_j	1	2	2	4	2	8	4	22
c_j	1	0	0	0	0	0	2	0

The nonzero c_7 is an explicit warning that the outer factor K cannot be discarded in an all-orders formula. The companion checks these counts and obtains

$$(g_0, \dots, g_6) = \left(1, -8, 22, -\frac{230}{3}, \frac{382}{3}, -\frac{22468}{15}, -\frac{283418}{45}\right).$$

The finite-polynomial and direct Q_J routes agree. A separate finite structural test compared the fiber product against full permutation fibers on all 3,004 asymmetric $R = 2$ constructions through order eight, finding no discrepancy or construction collision. That test is additional evidence for the finite implementation; the all-orders claim rests on the bijection and error bounds above.

9.3 The exact sign split and the total variation series

A class with $f(G) < 4$ has a realizer whose Klein orbit has fewer than four elements, hence a nontrivial stabilizer. Selecting one such realizer per class and using Lemma 5.1 bounds the number of these classes by $\exp((n/2) \log n + O(n))$. Their total contribution to either sampling law is smaller than every inverse power of n .

The first correction in (1.4) implies $4 < n!/a_n < 5$ eventually. Since fiber sizes are integers, $P_n(G) - U_n(G)$ is negative exactly for $f(G) \leq 4$. Total variation is the negative mass with its sign reversed. Write $A_n = 4a_n/n!$ and $B_n = 4M_{4,n}/n!$. For every fixed j ,

$$d_{\text{TV}}(P_n, U_n) = M_{4,n} \left(\frac{1}{a_n} - \frac{4}{n!} \right) + o(n^{-j}) = B_n(A_n^{-1} - 1) + o(n^{-j}). \quad (9.10)$$

This uses only the integer sign threshold, not divisibility of all fiber sizes by four. The latter is false: finite fibers of sizes three, five, and seven occur.

Define the ordinary-power version of the H_4 series by

$$V(y) = \sum_{k \geq 0} g_k \frac{y^k}{\prod_{j=0}^{k-1} (1 - jy)}. \quad (9.11)$$

Then A_n and B_n have the fixed-order inverse-power expansions $U(1/n)$ and $V(1/n)$, respectively. Since $U(0) = 1$, finite series inversion in (9.10) proves the following.

Theorem 9.3 (All-orders total variation). *Put*

$$Z(y) = V(y)(U(y)^{-1} - 1) = \sum_{k \geq 1} z_k y^k. \quad (9.12)$$

For every fixed integer $q \geq 1$,

$$d_{\text{TV}}(P_n, U_n) = \sum_{k=1}^q z_k n^{-k} + O_q(n^{-q-1}). \quad (9.13)$$

In particular,

$$d_{\text{TV}}(P_n, U_n) = \frac{4}{n} - \frac{15}{n^2} + \frac{169}{3n^3} + \frac{215}{2n^4} + O(n^{-5}).$$

To check the second coefficient alone, use $A_n^{-1} - 1 = 4/n + 17/n^2 + O(n^{-3})$ and $B_n = 1 - 8/n + O(n^{-2})$. Their product gives $4/n - 15/n^2$. The uncertainty of order n^{-2} in B_n is multiplied by $O(n^{-1})$, so it cannot affect this second coefficient.

The complete finite input tables displayed above give

k	g_k when available	z_k
0	1	0
1	-8	4
2	22	-15
3	-230/3	169/3
4	382/3	215/2
5	-22468/15	26449/15
6	-283418/45	232253/10
7	—	101570743/315

There is no missing input for z_7 : the series $U^{-1} - 1$ starts in degree one, so $g_0, \dots, g_6$ together with $h_0, \dots, h_7$ suffice. The seventh-order truncation therefore has $O(n^{-8})$ error at fixed order.

9.4 Scope of probabilistic transfer

For any sequence of graph events E_n ,

$$|P_n(E_n) - U_n(E_n)| \leq \frac{4}{n} + O(n^{-2}).$$

For real graph statistics Z_n with a uniform absolute bound M ,

$$|\mathbb{E}_{P_n} Z_n - \mathbb{E}_{U_n} Z_n| \leq 2M d_{\text{TV}}(P_n, U_n) = \frac{8M}{n} + O_M(n^{-2}).$$

If $0 \leq Z_n \leq 1$, the factor two can be omitted. An unbounded statistic requires additional tail or uniform-integrability control. The estimate is additive: an event of probability much smaller than $1/n$ need not have similar relative probabilities under the two laws. As with the enumeration series, no growing-order truncation is justified by the all-orders statement.

10 Source overlap and the boundary of the contribution

10.1 Exact enumeration already available in 1990

Bayoumi, El-Zahar, and Khamis [2] give prime-automorphism information, Pólya substitution, and a recursive generating-function system. Their Lemmas 4.2 and 4.4 and Proposition 4.5 are prior art for exact enumeration. Their Table 3, printed p. 4.47, agrees with OEIS A123448 through order fourteen and continues through order twenty. The original author-uploaded paper and the table page were inspected; no asymptotic enumeration statement was located there.

For bibliographic completeness, the six table entries beyond the inspected OEIS list are recorded below. They are *quoted prior-art table values*, not new enumerations certified by this report's companion.

n	a_n as printed in the 1990 table
15	232072611578
16	3819032840830
17	66512264611298
18	1222446378826186
19	23650094948059080
20	480517922278457424

The authors state that they used double precision. The remote table’s large-number typography and that numerical qualification are reasons not to describe all its high-order entries as newly verified exact integers. None of these six values enters the proof or the eight reported asymptotic coefficients. Exact finite verification here is bounded by order nine.

[write] **The quoted values and the expansion, 5 October 2026.** Two observations, neither a check of the table’s exactness nor of the theorem. (i) The values for $n = 15, \dots, 19$ are representable in IEEE double precision; the value for $n = 20$ is not: it lies in $[2^{58}, 2^{59})$, where doubles are multiples of 64, and it is $\equiv 16 \pmod{64}$. So that entry is not an IEEE 754 binary64 value. It is, however, exactly representable in the 56-bit-significand double formats in use in 1990 (IBM System/370 hexadecimal long, spacing 16 at this size; VAX D_floating, spacing 8), since $a_{20} \equiv 0 \pmod{16}$; the observation does not decide how the table was computed. (Corrected after the independent check of 5 October 2026; the first wording read “So that entry is not, as printed, the output of a plain double-precision computation”, which holds only for IEEE binary64.) (ii) With $S_8(n) = \frac{1}{4} \sum_{k=0}^7 h_k(n-k)!$, the relative difference $(a_n - S_8(n))/a_n$, computed from the OEIS terms ($n \leq 14$) and the quoted values ($n \geq 15$), is 0.1152, 0.0238, 0.0107, 0.00458, 0.00174, 0.000463, -0.0000604 and -0.000238 at $n = 12, 14, 15, 16, 17, 18, 19, 20$. The normalized differences $(a_n - \frac{1}{4} \sum_{k < R} h_k(n-k)!)/(n-R)!$ at $n = 20$ are -1.050 , -0.947 , -12.54 , -80.09 and -768.7 for $R = 1, \dots, 5$; their limits as $n \rightarrow \infty$ are $h_R/4 = -1$, -0.25 , -7.83 , -32.04 and -332.8 . The approach is slow and nothing here bounds a remainder.

10.2 Classical related asymptotics and sampling

The publisher abstract of El-Zahar–Sauer [6] states that the number of pairwise nonisomorphic two-dimensional posets on n elements is asymptotic to $n!/2$, using structural decomposition and unique representation. This is a close classical antecedent to the present leading equivalent. It counts posets rather than undirected comparability graphs; nevertheless a first-ever claim for $n!/4$ would require inspection of the complete proof and its consequences. The full subscription article was not obtained in this review.

Winkler’s paper [7], dated December 1990 by the publisher and listed as 1991 in an author bibliography, compares labeled, unlabeled, and random-linear-order-pair models of dimension-two orders. Its abstract reports the labeled equivalent $(n!)^2/(2\sqrt{e})$ and discusses rigidity, unique realizability, and uniquely orientable comparability graphs. Its full subscription text was likewise not inspected. These are sampling precedents, so (1.5) is presented as a concrete sharp quantitative refinement for the two explicitly defined graph laws, not as the first comparison of related random-order models.

Literal transitive orientations on labeled vertices must be separated from orientations modulo graph automorphisms. For example, flipping twin-edge vertices can change a labeled orientation while preserving an unlabeled poset type. The exact weighted identity in Section 2.2 avoids conflating these notions.

10.3 Later permutation and intersection graph sources

Johnston’s 2020 account [9] reports graph enumeration through order fourteen and says its author did not know whether the count was $o(n!)$. That is a historical statement about that account, not a comprehensive assessment of the literature.

[write] **Johnston’s question, 5 October 2026.** Johnston’s post [9], “Enumerating permutation graphs”, is dated 25 October 2020; its address is

<https://tomjohnston.co.uk/blog/2020-10-25-enumerating-permutation-graphs.html>.

It reports a computer enumeration through $n = 14$ (the source of OEIS a_{12} – a_{14}) and says “I don’t know if the number of permutation graphs is $o(n!)$ anyway”. Theorem 1.1 with $R = 1$ gives $a_n \sim n!/4$, so the count is not $o(n!)$. The qualitative answer was, however, already in print in the sources read: the proof of Theorem 2.1 of arXiv:2402.06394v1 (9 February 2024, Section 4.3) shows $a_n \geq n!/30$ for all large n , which is the coarse lower bound quoted in the next paragraph. Their argument (every simple permutation realizes a modular-prime graph, and such a graph has at most four realizers) gives $a_n \geq s_n/4$, hence $\liminf_n a_n/n! \geq e^{-2}/4 \approx 0.0338$, and $1/30$ is a convenient weakening of that, since $4e^2 \approx 29.56 < 30$. Relative to these two sources, the present report adds the constant $1/4$ and the expansion to every fixed order; whether El-Zahar–Sauer [6] already implies the constant is Question 6 of Section 12.1. (Wording qualified after the independent check of 5 October 2026; the first wording read “ $1/30$ is a rounding of that” and “What the present report adds is the constant $1/4$ and the expansion to every fixed order”, which read as a priority claim that the El-Zahar–Sauer caveat leaves open.) The placement record of batch 106 said that this report answers Johnston’s question; that holds for the sharp constant, not for the dichotomy itself.

The inspected arXiv:2402.06394v1 (2024) of Bassino–Bouvel–Féray–Gerin–Pierrot [5] studies scaling and graphon limits and uses coarse bounds $n!/30 \leq a_n \leq n!$ in its permutation-graph discussion. A separately available author-hosted PDF has February 2024 file metadata and the same coarse lower bound; it is not evidence that the final journal text has been inspected.

The final article, *Dense and nondense limits for uniform random intersection graphs*, appeared in *Electronic Journal of Probability* 31 (2026), article 110, DOI 10.1214/26-EJP1570. The official landing and PDF requests returned an access-error page in the source review. The final text remains uninspected, so possible overlap with it is explicitly unresolved. No access-control bypass, author contact, or purchase was used.

10.4 What is established here

The graph-specific result is the finite rooted-context and large-prime-quotient approximation (5.4), followed by its all-orders coefficient formula and the stated inverse and sampling consequences. Simple-permutation decomposition, its exact functional equation, and its full factorial asymptotics are established inputs. The current report and repository-index search found no matching permutation-graph topic, but this was a bounded absence check, not a universal novelty proof.

[write] **5 October 2026.** That search was true when it was made: before batch 106 no file of the repository treated A123448 or the enumeration of permutation graphs. This report is now that topic, filed with two siblings that share its pipeline (Section 1.1).

The conclusions that require no unresolved priority assertion are: the proof gives the fixed-order expansion in the stated model; the eight coefficients have independent exact checks; the inverse windows preserve rounding; and the sharp total-variation constant is four for the laws (9.1). No OEIS submission, external publication, or public repository modification is part of this report.

11 Reproducibility and verification limits

The source archive contains this LaTeX source, the rendered PDF, standard-library Python exact-checking programs and tests, machine-readable coefficient claims, a source-provenance record, checksums, and deterministic release scripts. It contains neither unrelated earlier reports nor private review documents. The root README supplies the exact commands and expected outputs.

The fast companion replay checks rational series identities, coefficient agreement by two routes, fixed-cutoff stabilization, ordinary-power conversion, logarithmic coefficients, the symbolic second-shift cancellation, and denominator divisibility. Exhaustive graph replay is separately selectable: small orders provide a fast regression test, while order nine reruns all finite inputs used for the displayed eight coefficients. A full dual-route order-nine replay is materially slower and is deliberately not hidden inside the default tests.

Tests run under both ordinary Python and `python3 -0`; verification does not depend on the presence of Python `assert` statements. Command-line outputs use new files or new directories, reject existing destinations and symbolic-link traversal, and avoid silently overwriting the report or a prior result. Release manifests use a fixed allowlist, and the ZIP has fixed member order, timestamps, and metadata. PDF rebuilding uses fixed metadata, no shell escape, a clean TeX environment, and reference passes to a stable auxiliary state. Byte-for-byte reproduction is expected within the recorded toolchain; another TeX or font version may typeset identically without producing identical bytes.

The release checks certify the finite calculations and packaging, not the analytic proof by executable means. In particular they do not establish a numerical onset for the asymptotic inequalities, turn the formal generating series into convergent analytic functions, validate a growing truncation order, or resolve the uninspected-source priority questions. Those boundaries are part of the result rather than omissions that a finite successful test can erase.

12 Further research questions

Effective remainder constants and threshold decisions. The interval-tail proof can be made numerical at each fixed order, but an effective final error also requires explicit constants in the simple-permutation transfer and in its finite-polynomial composition. A useful next goal is to propagate all of these bounds together, producing specified pairs (K_R, n_R) for (8.7). This would turn the asymptotic threshold window into a certified large-input test: evaluate its two integer endpoints, and, when they differ, settle the remaining case by an independent exact bound on a_n . Such a certificate would need directed numerical error control in the gamma and Lambert- W evaluations as well as the asymptotic remainder; floating-point agreement alone is insufficient.

Coefficient growth and orders that increase with size. The rational-coefficient formal generator (1.2) determines every fixed coefficient, but the growth of h_k is not bounded in this report. Additional exact inputs could guide a study of coefficient growth, while the proof would need estimates uniform in the cutoff R : the number of bounded decorations, the interval-exclusion constant, and the factorial-transfer remainder all change with R . Until that dependence is controlled, neither a truncation order tending to infinity nor an optimal truncation rule is justified. Questions about Borel summation and exponentially small terms go further. The symmetric-quotient estimate isolates one negligible scale, but does not classify all sectors beyond the factorial Poincaré expansion

or their interactions. A complete transseries description would require new information beyond the fixed-order theorem.

Efficient finite input recurrences. Exhaustive canonicalization makes the coefficient formula effective at every finite order but rapidly becomes expensive. The 1990 Pólya recurrence offers an established starting point for a faster unrooted calculation. Extending exact recursive enumeration to vertex-rooted contexts and, separately, to one-realizer graphs and singleton root orbits could make many more h_k , g_k , and z_k accessible. The marked multiplicity (9.2) is the quantity that must be retained: ordinary vertex pointing by na_n does not supply r_n , and dropping nontrivial marked contexts fails already for seven-vertex contexts. A practical recurrence should first be reconciled with both exhaustive routes on their overlapping range.

Finer fiber laws and sampling comparisons. Lemma 9.1 keeps more information than the event $f(G) = 4$: it factors the full fiber through a marked context and all port fibers. One can therefore investigate other fixed fiber sizes or bounded functions of fiber size by retaining these factors rather than only selecting those equal to one. The principal technical questions are how to organize the resulting finite input data and how to control the weighted exceptional classes. For unbounded positive weights, the existing class-count error is not enough. Likewise, relative comparison of rare events or expectations of unbounded graph statistics needs event-specific or tail estimates, not merely the total-variation series proved here.

12.1 Further questions and research

[write] **Added 5 October 2026, batch 106.** Under the repository’s standing rule (Vladimir Reshetnikov, 4 October 2026), every claim of the source that is not proved here is stated below as an open question, with its source, the sketch it came with and what is missing. Nothing in the source was found to be wrong. One reading of the placement record is qualified: the source settles the constant of Johnston’s question, not the dichotomy, which was already in [5] (note in Section 10).

1. *Effective constants and certified thresholds* (source, first paragraph above; Theorem 8.1 and (1.7)). Find explicit pairs (K_R, n_R) for (8.7), and an explicit K and onset for (1.7), by propagating the constants of the interval tail (Lemma 3.1), the symmetry bound (Lemma 5.1) and the factorial transfer [3]. Missing: an effective form of Borinsky’s composition theorem and directed error control in the gamma and Lambert- W evaluations. Near a jump, two integers remain possible whatever the precision (Remark 8.2 (a)).
2. *Explicit centres of every order* (source, Section 8: “the coefficients and formal shifts can be computed to any fixed order”). The formal shifts are the coefficients e_n of Remark 8.2 (d), so they exist and are computable. A bracket of width $O(u^{-m}/D)$ around the centre truncated after m shifts is proved here only for $m = 2$, in (1.7); the general case should follow from Theorem 8.1 and an expansion of ν_R , but is not written out.
3. *Growth of h_k and orders that increase with n* (source, second paragraph above, and Remark 6.2). Bound h_k (the write’s $h_8 = -3687969659/2520$, from the OEIS term a_{10} , confirmed at the independent check, continues the growth of $|h_k|$), make the constants uniform in R , and decide whether a truncation order $R(n) \rightarrow \infty$, optimal truncation, Borel summation or a description of the exponentially small sectors is justified. Missing: every estimate uniform in R .
4. *Efficient finite inputs* (source, third paragraph above). Extend the 1990 Pólya recurrence [2] to the vertex-rooted counts r_n , the one-realizer counts b_j and the marked contexts c_j , and reconcile it with both exhaustive routes. Each new pair of terms gives one more coefficient: h_9 needs r_{10} (a_{11} is in the OEIS), g_7 needs b_9 , and z_8 needs g_7 and h_8 .

5. *Finer fiber laws and sampling* (source, fourth paragraph above, and the scope statements of Section 9). Expand the laws of other fixed fiber sizes through Lemma 9.1; compare rare events relatively and unbounded statistics. Missing: control of weighted exceptional classes and tail estimates; the class-count error does not suffice for unbounded weights.
6. *Priority and the quoted table* (source, “Scope and priority” and Section 10). Does the full text of El-Zahar–Sauer [6] (two-dimensional posets $\sim n!/2$) or of Winkler [7] contain $a_n \sim n!/4$ or its method; what does the final journal text of [5] (2026) say about permutation graphs; are the 1990 values for $n = 15, \dots, 20$ exact (the printed a_{20} is not an IEEE double, note in Section 10; it is exact in the 56-bit double formats of 1990)? Missing: the three full texts and an exact enumeration beyond $n = 14$.

[write] **Independent check of the write, 5 October 2026.** An adversarial check made by the intake after the write (commit `6fb3059e5`) reread Johnston’s post and OEIS A123448 (revision 35; a_{11} by Jablonski, a_{12} – a_{14} by Johnston) and a fresh download of arXiv:2402.06394v1 (Proposition 4.6 and the proof of Theorem 2.1 in Section 4.3, with “using $4e^2 < 30$ ”), and confirmed the account of both in Section 10. It confirmed the four parts of Remark 8.2 against the transseries volume: the staircase hypotheses with $n_0 = n_1 = 1$, that the bracket is not an instance of `p0:thm:staircase(1)–(2)`, the factorial core with $\kappa = 1$, $d = -1$, and the two shifts as e_1, e_2 of `p0:thm:core-reversion` (with $q_1 = -47/12$, $q_2 = -9$ recomputed). By its own exhaustive enumeration (canonical forms of the inversion graphs of all permutations, self-tested on the graph counts 1, 2, 4, 11, 34, 156 of A000088) it found $a_1, \dots, a_{10}$ and $r_1, \dots, r_9$ as in the table and the OEIS, in particular $a_{10} = 524,572$, and its own exact series code gave $h_0, \dots, h_8$, $8! h_8 = -59,007,514,544$ and $[x^{10}]Q = -1,319,072$; the caveat of the note on h_8 is discharged there. It confirmed that $a_{15}, \dots, a_{19}$ are IEEE doubles and a_{20} is not, and every number of the note on the quoted values. One inference there was wrong as worded and is corrected with a dated note keeping the first wording: a_{20} is not an IEEE binary64 value, but it is divisible by 16 and so exact in the 56-bit double formats of 1990 (IBM System/370 hexadecimal long, VAX D_floating), so the observation does not show that the table was not computed in double precision. Two wordings of the note on Johnston’s question were qualified, also with the first wording kept: the report adds the constant $1/4$ relative to the two sources discussed there (Question 6), and $1/30$ is a weakening of $e^{-2}/4$ rather than a rounding. No mathematical error was found.

References

- [1] M. H. Albert, M. D. Atkinson, and M. Klazar, *The enumeration of simple permutations*, Journal of Integer Sequences **6** (2003), Article 03.4.4. Journal article and full text.
- [2] B. I. Bayoumi, M. H. El-Zahar, and S. M. Khamis, *Counting Permutation Graphs*, 25th Annual Conference on Statistics, Computer Science and Operations Research, Cairo University, 8–10 December 1990, pp. 4.23–4.49. Institutional record; author-uploaded full text.
- [3] M. Borinsky, *Generating asymptotics for factorially divergent sequences*, Electronic Journal of Combinatorics **25**(4) (2018), P4.1. Published full text. Theorem 35 and equations (62)–(67).
- [4] M. Habib and C. Paul, *A survey of the algorithmic aspects of modular decomposition*, Computer Science Review **4**(1) (2010), 41–59. Author-hosted final text. Lemma 20, printed p. 55.
- [5] F. Bassino, M. Bouvel, V. Féray, L. Gerin, and A. Pierrot, *Dense and nondense limits for uniform random intersection graphs*, arXiv:2402.06394v1 (2024). Exact inspected version. The associated final article is *Dense and nondense limits for uniform random intersection graphs*, Electronic Journal of Probability **31** (2026), article 110, 44 pp. Final journal record. The final text was not inspected.

- [6] M. El-Zahar and N. W. Sauer, *Asymptotic enumeration of two-dimensional posets*, Order **5** (1988), 239–244. Publisher abstract and record. Full subscription text not inspected.
- [7] P. Winkler, *Random orders of dimension 2*, Order **7**, 329–339 (publisher date December 1990; author bibliography 1991). Publisher abstract and record. Full subscription text not inspected.
- [8] The OEIS Foundation, *A123448 Number of permutation graphs on n vertices*. OEIS entry. Inspected source snapshot, 3 October 2026.
- [9] T. Johnston, *Enumerating permutation graphs*, 25 October 2020. Author’s enumeration account.
- [10] [write] ProveIt, *Transseries and inversion*, the transseries-and-inversion volume
`(p0:prop:factorial-core, p0:thm:core-reversion, p0:rem:core-instances,`
`p0:def:three-inverses, p0:thm:staircase),`
`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`
`transseries_and_inversion.tex.`